# Structural Limits: A Unified Theory of Collapse Across Domains


## Abstract

Across scientific disciplines, systems occasionally undergo abrupt transitions where their internal dynamics cease to operate. Cosmological singularities, biological origin events, financial crises, technological failures, and paradigm shifts in science all share a common feature: the breakdown of previously valid dynamical descriptions. Despite extensive study in individual fields, no general theory explains why such collapses occur across domains.

This paper proposes a unified framework based on **structural limits of dynamical derivability**. We argue that every dynamical system operates within a structurally admissible domain defined by conditions that are not generated by the dynamics itself. When system trajectories approach the boundary of this domain, the governing dynamics cannot generally be extended while preserving internal consistency. Collapse events therefore correspond not to extreme behavior of the dynamics but to encounters with **structural boundaries of derivability**.

The proposed framework separates **dynamical evolution** from **structural admissibility**, introducing a general theory of collapse applicable across physics, biology, economics, cognition, and technological systems. By formalizing structural limits, the theory explains why catastrophic transitions are often retrospectively intelligible yet intrinsically difficult to predict from internal dynamics alone. The framework unifies a wide range of collapse phenomena within a single conceptual architecture and provides a foundation for cross-domain analysis of structural transitions.


## 1. Introduction

Many of the most consequential events in complex systems take the form of sudden transitions. Cosmological models encounter singularities where known physical laws cease to apply. Biological processes exhibit origin events—such as the emergence of life or individual conception—that cannot be derived from prior system dynamics alone. Financial systems periodically experience crises in which previously stable mechanisms fail abruptly. Scientific disciplines themselves undergo paradigm shifts in which the conceptual framework of a field collapses and is replaced.

Despite occurring in very different domains, these events share a striking structural similarity: **the governing dynamics cease to be valid**.

Traditional approaches attempt to interpret such events as extreme cases of dynamical behavior. In catastrophe theory, for example, sudden transitions arise from continuous changes in control parameters. In nonlinear dynamics, bifurcations mark qualitative changes in system behavior. In thermodynamics, far-from-equilibrium processes produce emergent structures.

However, these approaches typically assume that the **dynamical laws themselves remain valid across the transition**.

This paper explores a different possibility: that collapse events occur when systems encounter **limits of structural admissibility** beyond which the internal dynamics cannot be defined.

Under this perspective, collapse does not represent unusual dynamical behavior but rather the **failure of the domain within which the dynamics is defined**.

The central claim of this work is therefore simple but far-reaching:

> *Every dynamical system operates within a structurally admissible domain, and collapse occurs when system trajectories encounter the boundary of that domain.*

This idea leads to a general framework capable of unifying collapse phenomena across disciplines.

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## 2. Structural Domains of Dynamical Systems

Any dynamical system describes how states evolve over time according to some rule or operator.

In conventional formulations, attention focuses primarily on the properties of the dynamical rule itself: stability, attractors, bifurcations, and chaotic regimes.

Yet every dynamical description implicitly assumes the existence of a **domain in which the dynamics is defined**.

This domain includes not only the set of possible system states but also the structural conditions that allow the dynamical rules to operate. These conditions may involve physical constraints, biological organization, institutional frameworks, or conceptual structures depending on the domain.

Within this domain, the system evolves according to its internal dynamics.

Outside it, the dynamical description loses validity.

The structural domain therefore acts as a **boundary condition for dynamical possibility**.

Most analyses focus on behavior inside the domain. Structural limits, however, arise at its boundary.

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## 3. Structural Non-Derivability

A key consequence of structural domains is that their boundaries cannot, in general, be derived from the internal dynamics of the system.

The reason is straightforward. If the dynamics were capable of deriving the conditions of its own applicability, then the domain boundary would itself belong to the dynamical description. But in practice, the domain is defined by conditions that precede or constrain the dynamics.

As a result, when system trajectories approach the structural boundary, the governing dynamical rules may cease to operate.

Collapse events therefore correspond to **points at which the dynamical description becomes non-derivable**.

In such cases, no extension of the internal dynamics can continue the trajectory while preserving the same structural framework.

This leads to a general principle:

Collapse events correspond to encounters between system trajectories and structural boundaries beyond which the governing dynamics cannot be extended.

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## 4. Geometry of Structural Collapse

Structural limits can be understood geometrically.

Within the state space of a system, the structurally admissible region forms a domain inside which the dynamical rules apply. The boundary of this domain represents a structural limit of derivability.

As trajectories evolve, they may approach this boundary. When they reach it, the internal dynamical description fails.

Collapse therefore corresponds to the termination of dynamical trajectories at structural boundaries.

This geometric view makes it possible to analyze collapse phenomena in a unified manner across different systems.

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## 5. Cross-Domain Manifestations

The concept of structural limits appears in many disciplines, though often under different names.

In cosmology, singularities mark points where current physical laws cannot describe the state of the universe.

In biology, origin events such as conception represent transitions between organizational regimes that cannot be derived from the dynamics of the preceding state alone.

In economics, systemic crises occur when institutional structures that support market dynamics fail.

In science itself, paradigm shifts replace the conceptual frameworks that define acceptable explanations.

These phenomena share a common structure: the breakdown of the domain within which existing dynamics operates.

The theory of structural limits therefore provides a unified interpretation of collapse events across domains.

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## 6. Retrodictive Intelligibility

An important feature of structural collapse is that it is often easier to explain retrospectively than to predict in advance.

Before a collapse occurs, the internal dynamics may appear stable and well-behaved. Only after the event does it become clear that the system had approached a structural boundary.

This asymmetry explains why catastrophic transitions often seem surprising even when extensive data about the system is available.

The difficulty lies not in insufficient information but in the fact that the **structural boundary itself is not derivable from the internal dynamics**.

Understanding collapse therefore requires recognizing the limits of dynamical explanation.

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## 7. Implications for the Study of Complex Systems

The theory of structural limits has several implications for the study of complex systems.

First, it suggests that optimizing system performance within an existing structure does not guarantee long-term persistence. Systems may function efficiently while simultaneously approaching structural limits.

Second, it implies that catastrophic transitions cannot always be predicted from internal dynamics alone.

Third, it provides a conceptual framework capable of integrating collapse phenomena across scientific disciplines.

By distinguishing between dynamical evolution and structural admissibility, the framework clarifies why collapse is a recurring feature of complex systems.

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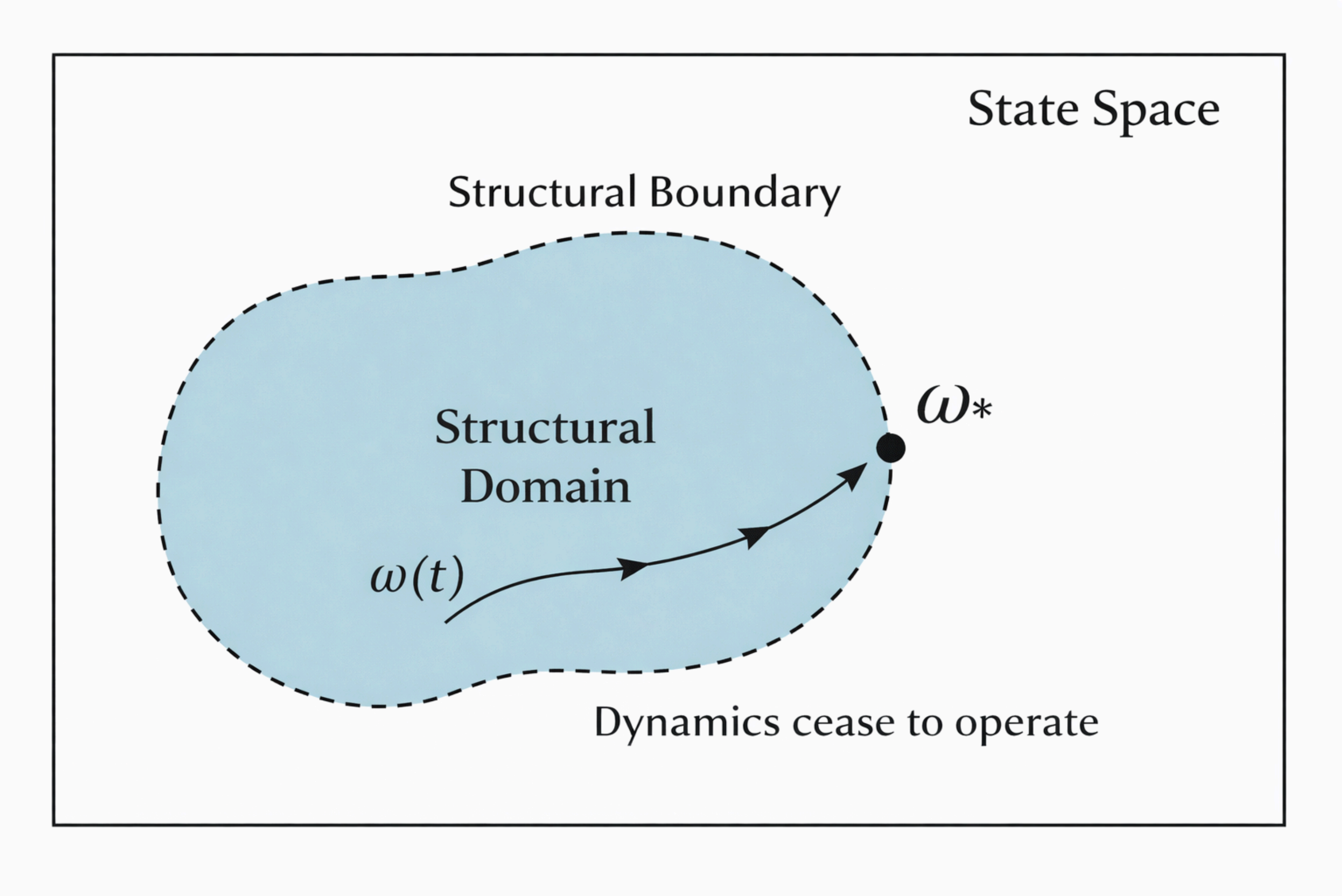
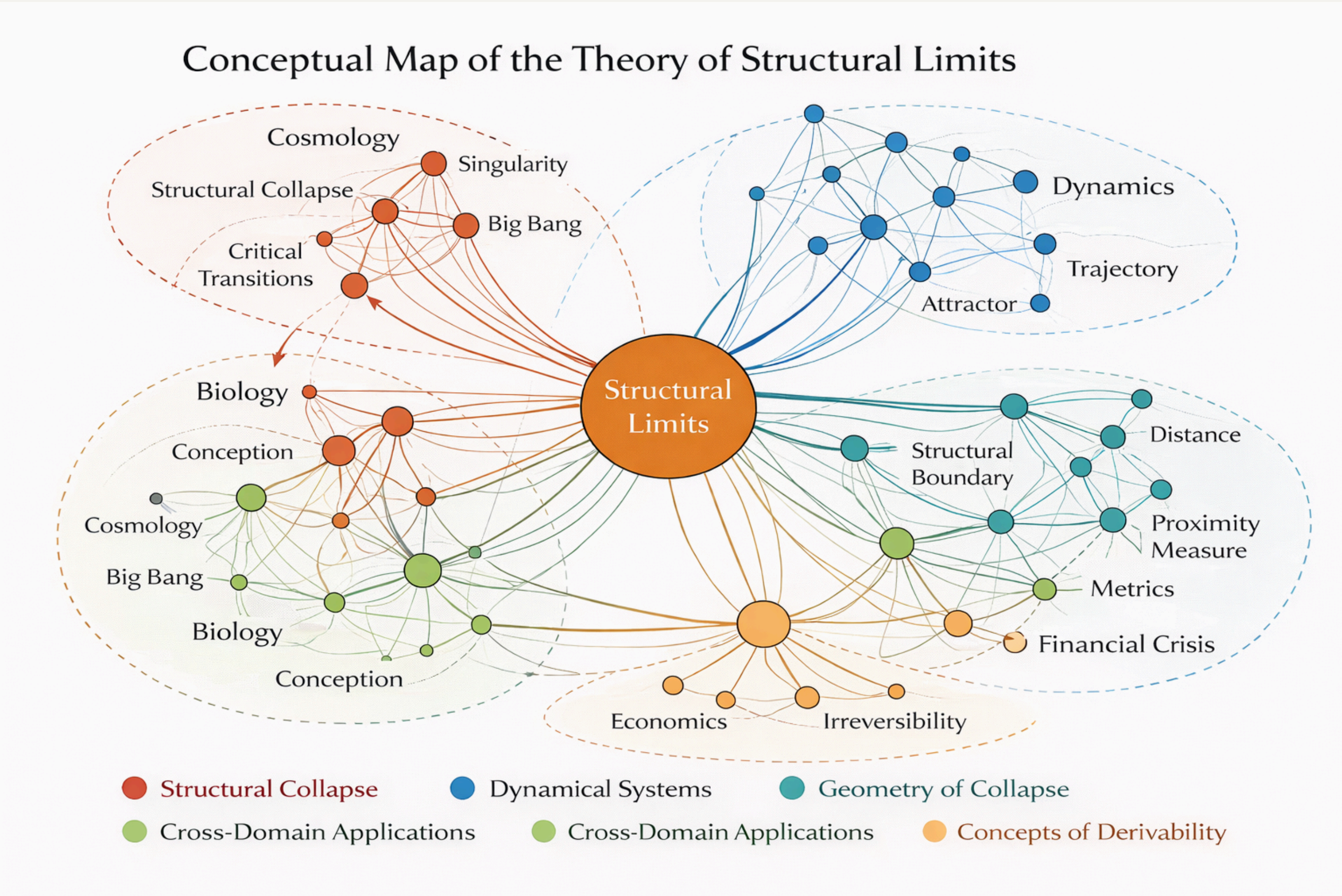
## 8. The Structural Limits Research Program

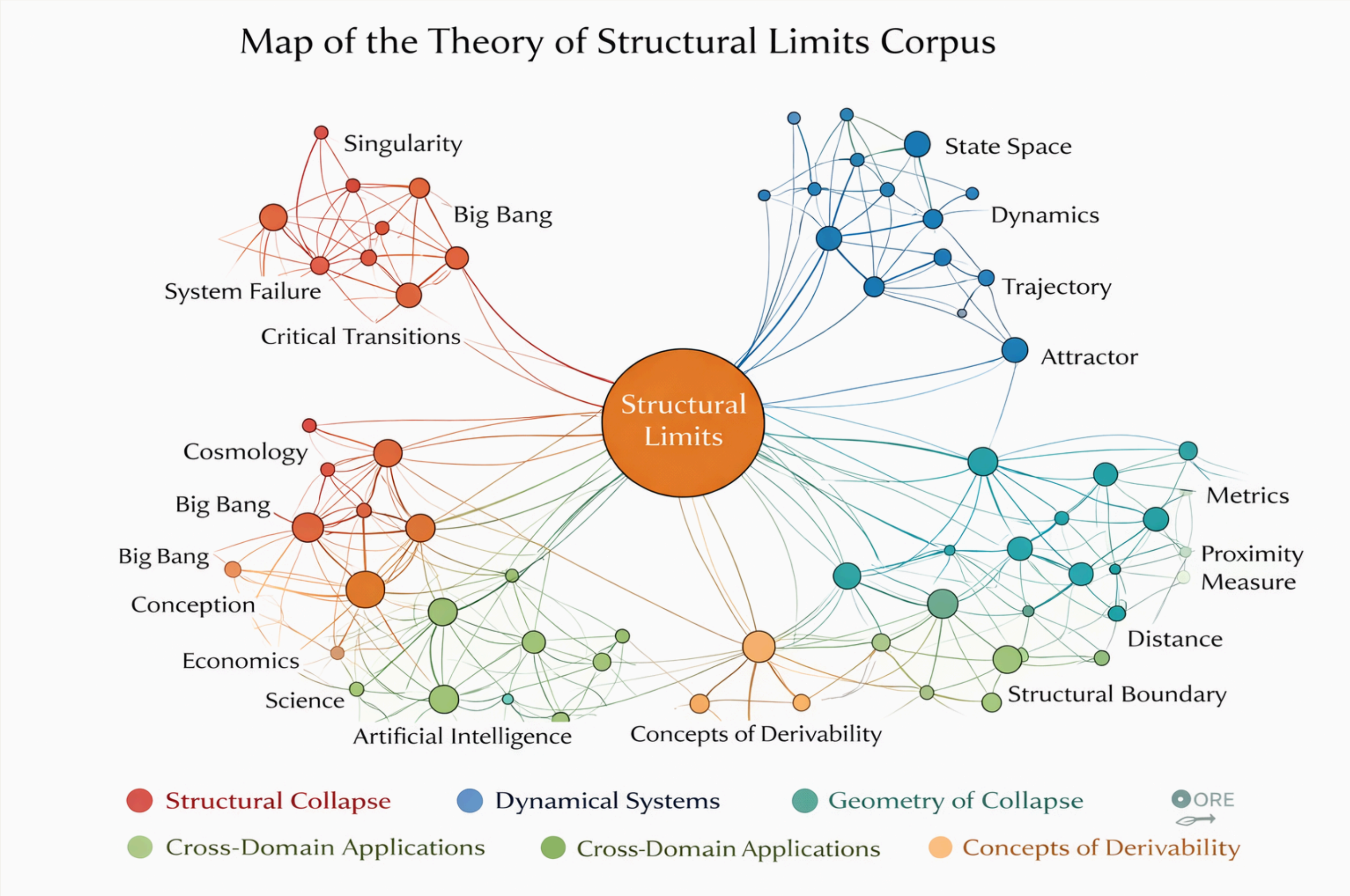
The ideas presented here form part of a broader research program exploring structural limits across multiple domains. The program includes theoretical work on structural boundaries, geometric measures of collapse proximity, and domain-specific applications in physics, biology, economics, and technological systems.

Together, these studies suggest that structural limits represent a fundamental organizing principle of complex systems.

## Structural Domain Diagram






## 9. Conclusion

Collapse events play a central role in the evolution of complex systems. Yet they have traditionally been studied within the boundaries of individual disciplines.

This paper proposes a unified perspective based on the concept of **structural limits of dynamical derivability**. According to this view, systems operate within structurally admissible domains defined by conditions that their internal dynamics cannot generate. When trajectories encounter the boundary of these domains, the governing dynamics ceases to apply.

Collapse therefore reflects not extreme dynamical behavior but the limits of the structures that make dynamical evolution possible.

Recognizing these limits opens a new path toward a general theory of structural transitions across domains.